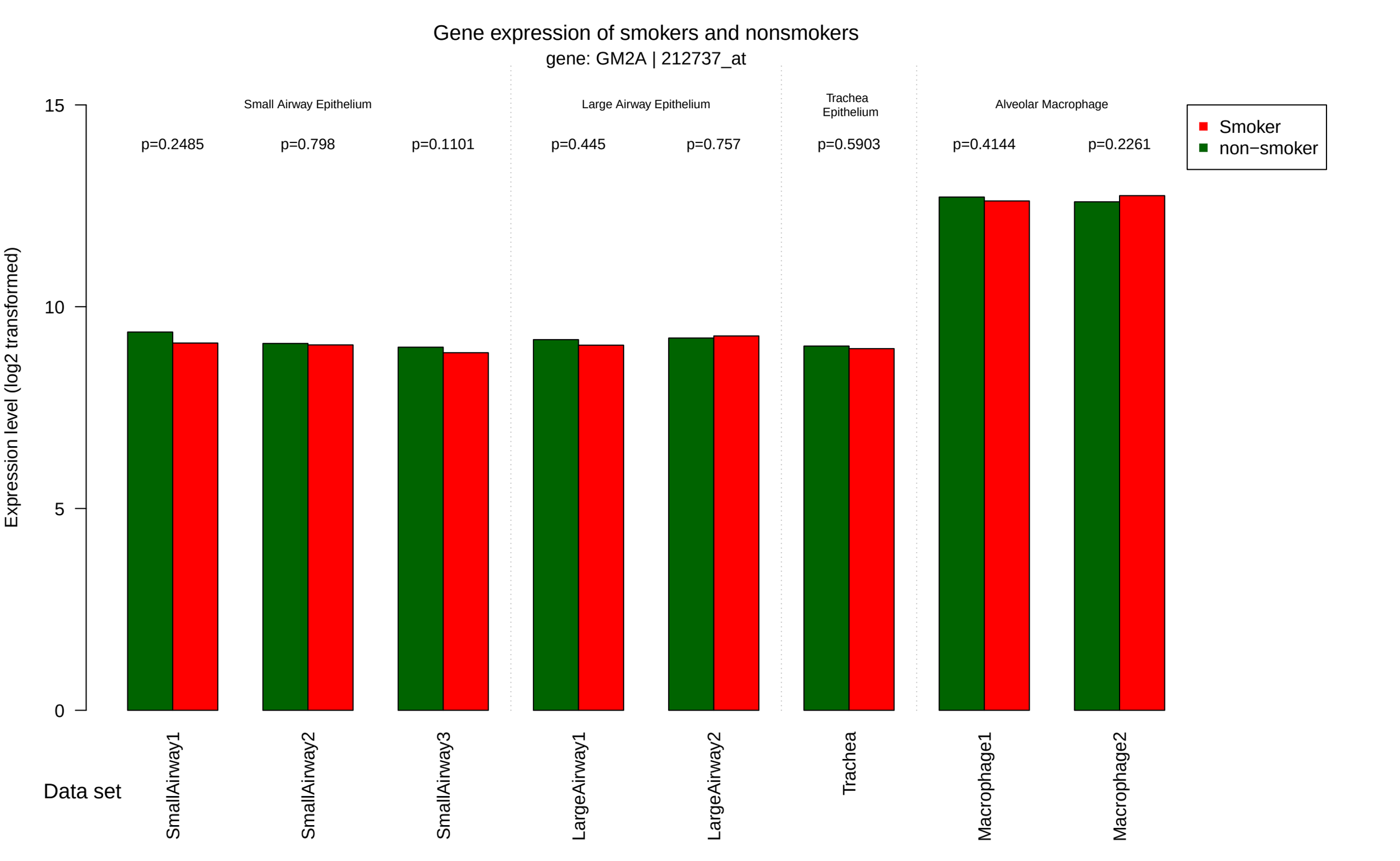
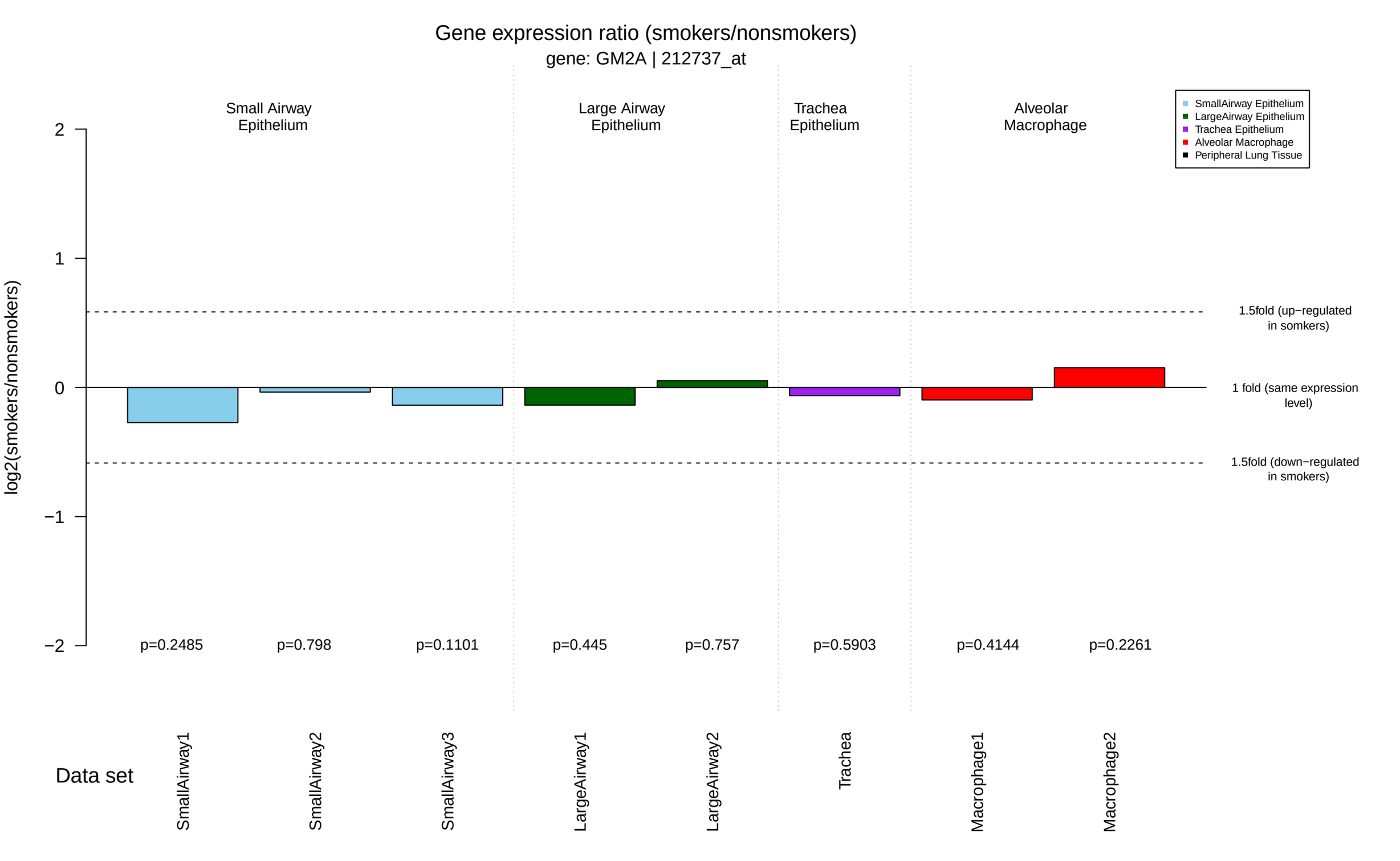
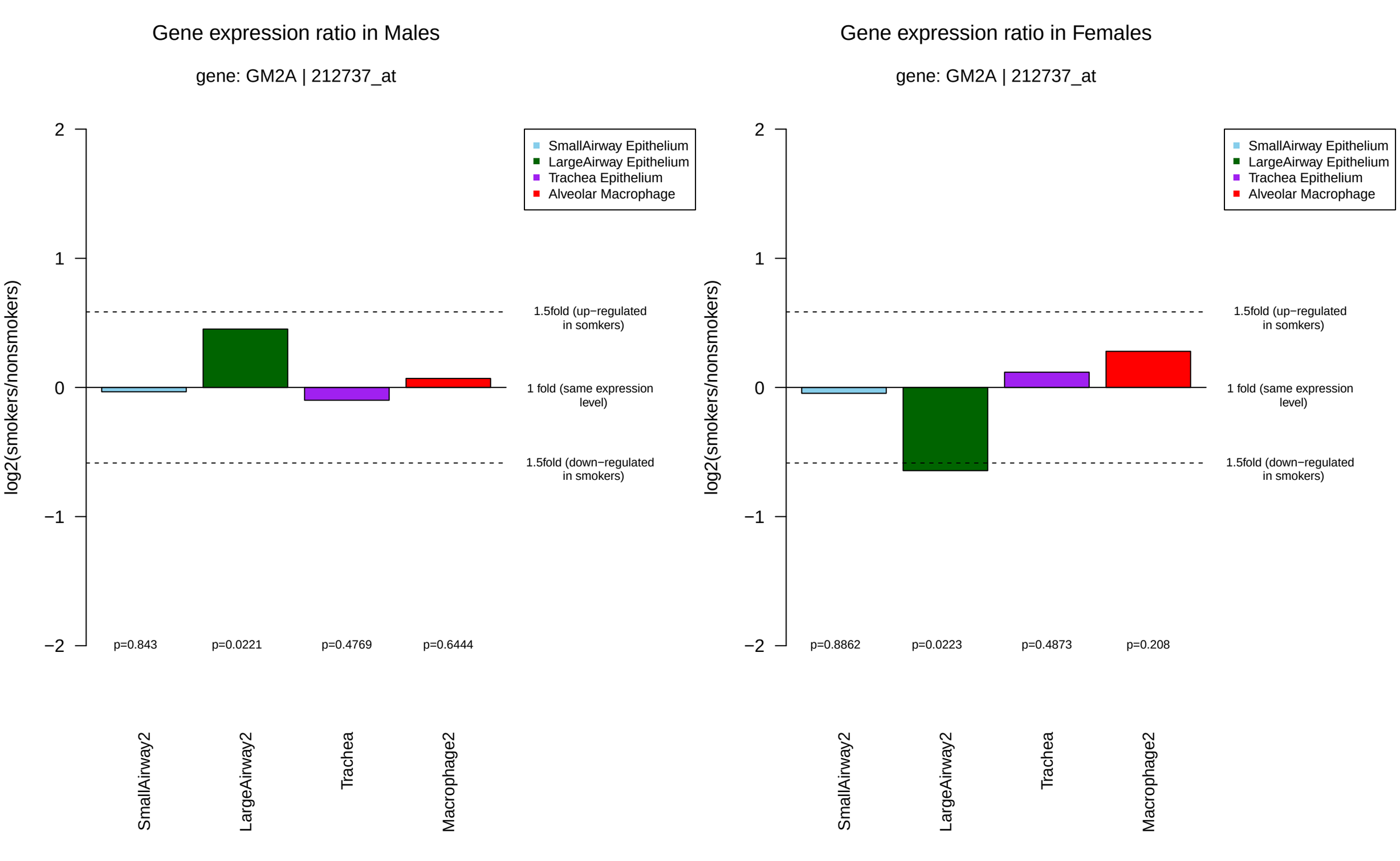
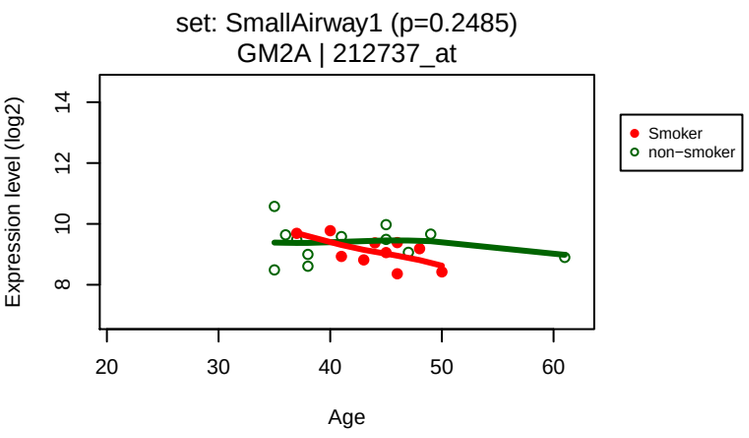
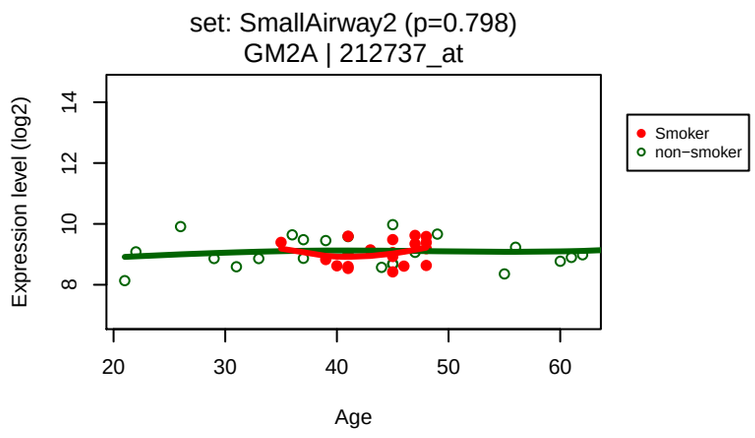
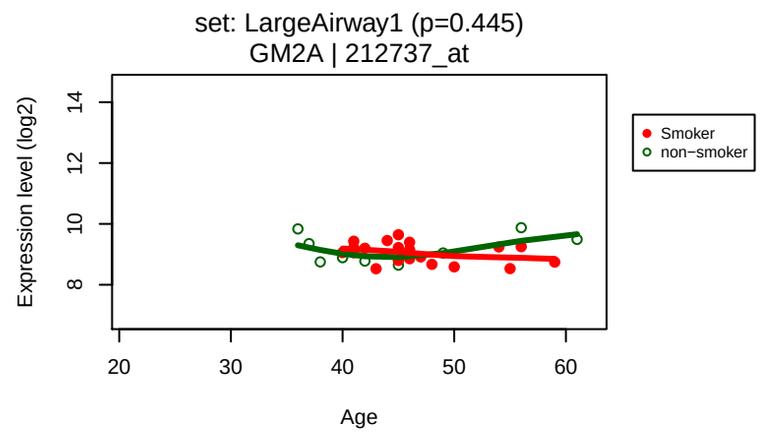

Gene expression of smokers and nonsmokers
gene: GM2A | 212737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2485
p=0.798
p=0.1101
p=0.445
p=0.757
p=0.5903
p=0.4144
p=0.2261
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

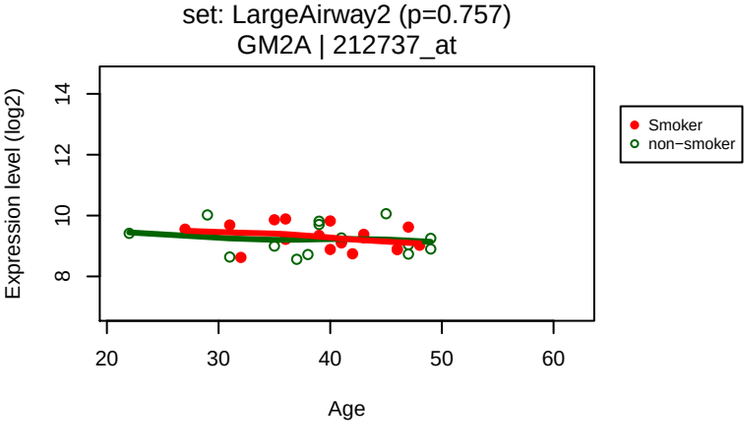

Gene expression ratio (smokers/nonsmokers)
gene: GM2A | 212737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2485
p=0.798
p=0.1101
p=0.445
p=0.757
p=0.5903
p=0.4144
p=0.2261
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

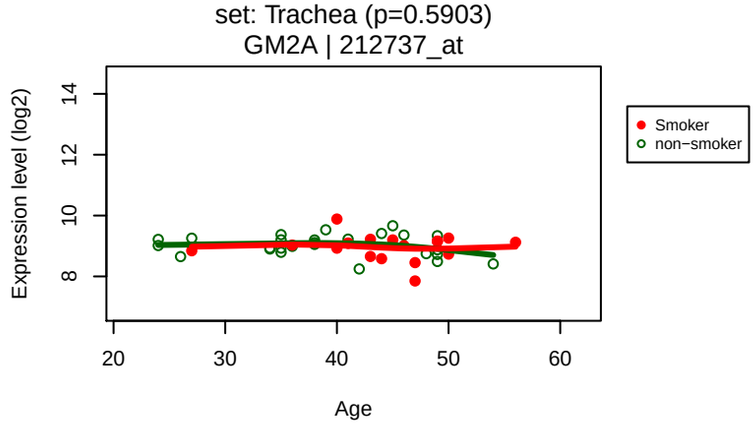

Gene expression ratio in Males
gene: GM2A | 212737_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.843
p=0.0221
p=0.4769
p=0.6444
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GM2A | 212737_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8862
p=0.0223
p=0.4873
p=0.208
SmallAirway2
LargeAirway2
Trachea
Macrophage2

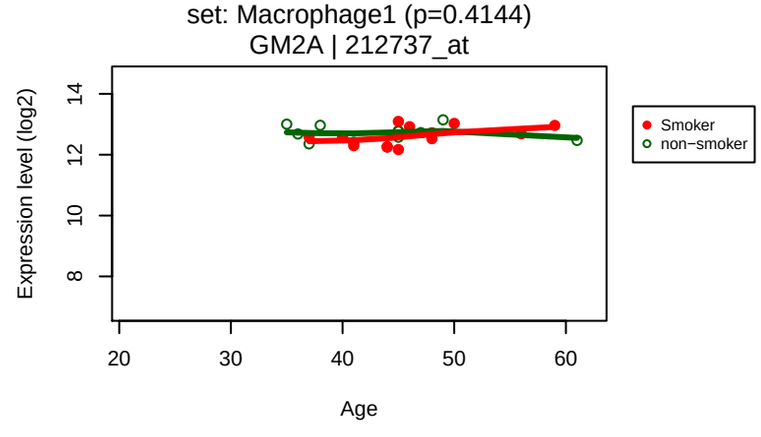

set: SmallAirway1 (p=0.2485)
GM2A | 212737_at
Expression level (log2)
Age
Smoker
non−smoker

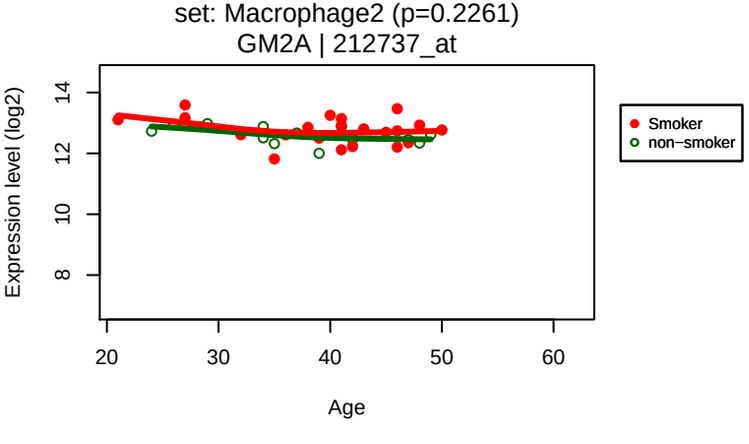

set: SmallAirway2 (p=0.798)
GM2A | 212737_at
Expression level (log2)
Age
Smoker
non−smoker

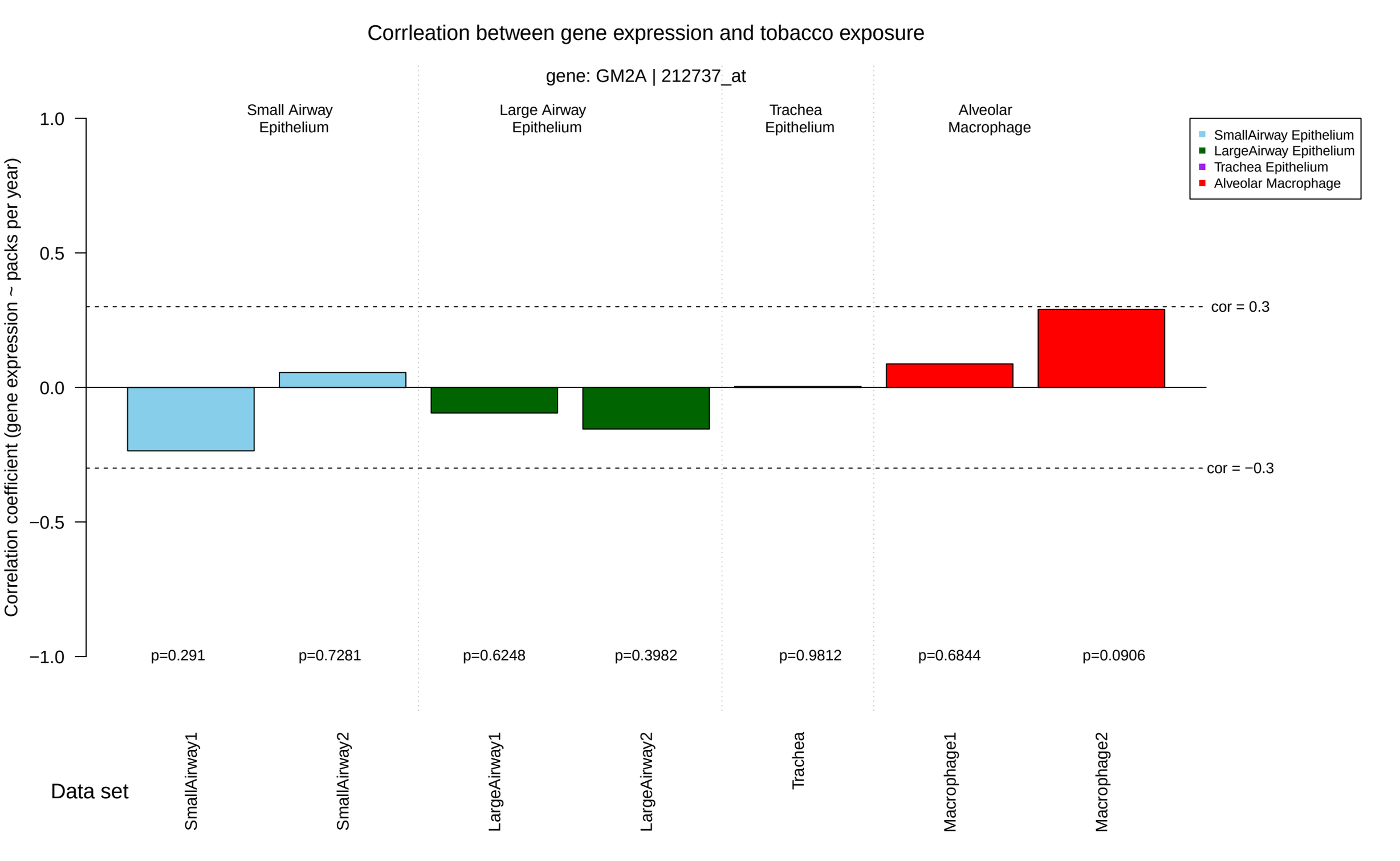

set: LargeAirway1 (p=0.445)
GM2A | 212737_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.757)
GM2A | 212737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5903)
GM2A | 212737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4144)
GM2A | 212737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2261)
GM2A | 212737_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GM2A | 212737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.291
p=0.7281
p=0.6248
p=0.3982
p=0.9812
p=0.6844
p=0.0906
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2